

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

Date: 10/01/2026	Improvement-CIE	Max. Marks:(10+50) = 60
Semester: VII	UG	Duration: 2 Hours
Course Title: Computer Graphics & Virtual Reality	Course Code: CS373IA	

Q. No.	QUIZ	M	BT	CO
1.1	Differentiate between image-based and object-based method for hidden surface removal algorithms?	2	L1	CO1
1.2	Differentiate between parallel projection and perspective projection.	2	L2	CO2
1.3	In area sub-division method state the conditions when further sub divisions are required.	1	L2	CO2
1.4	Identify which HSR algorithm provides entire cross section of 3D object into 8 regions.	1	L2	CO3
1.5	List the purpose of collision detection in physical modelling?	1	L3	CO5
1.6	Illustrate the role of haptic texturing?	1	L3	CO5
1.7	What does behavior modelling describe in virtual reality?	1	L2	CO4
1.8	What is a projection in which all three foreshortening factors are kept equal is called as?	1	L2	CO4

Q. No.	TEST	M	BT	CO
1.a	Explain the steps involved in the 3-D viewing pipeline.	5	L2	CO3
1.b	Discuss in detail the classification and working of projections in 3-D graphics.	5	L3	CO2
2.a.	Discuss Scanline line algorithm for detecting visible surface with diagram.	5	L2	CO2
2.b.	Analyze how BSP-tree approach is useful when multiple object views with multiple reference points are available.	5	L2	CO2
3.	Examine the stages of the haptics rendering pipeline and analyze how each stage contributes to the overall realism and usability of a physical modelling system?	10	L3	CO3
4.a	Analyze how the Level of Autonomy (LOA) influences the perception, decision-making, and action selection of autonomous agents in virtual environments. Support your answer using the example of a virtual office.	5	L4	CO5
4.b	Geometric modeling describes the shape of virtual objects and their appearances. Discuss different methods for modeling 3D object geometry.	5	L3	CO4
5.a	Analyze the importance of parent-child relationships in object hierarchies. How do these relationships enable both collective movement and independent motion of object parts? Support your answer using the example of a virtual football player.	7	L4	CO5
5.b	List and explain the different stages of VR Modeling cycle.	3	L2	CO4

Course Outcomes: After completing the course, the students will be able to: -

CO1:	Apply the basic concepts of Computer Graphics & Virtual Reality which illustrates the use of the pipeline architecture, OpenGL library, VR tools.
CO2:	Analyze and make an appropriate choice of methods required for computer representation of 2D/3D objects and modelling in Virtual Reality.
CO3:	Design Computer Graphics and Virtual Reality applications using OpenGL library & VR tools.
CO4:	Implement common geometric construction & VR techniques and develop Computer Graphics/ VR based solution to Engineering applications / real-world problems.
CO5:	Exhibit team work and effective oral/written communication skills to accomplish a common goal of solving Computer Graphics/Virtual Reality problems with the engineering community and society at large.